

Artificial Intelligence & Machine Learning

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Propositional Tableaux

How to Check Satisfiability

How to check whether there exists an interpretation I s.t. $I \models \varphi$?

We have seen Truth Tables: for complex formulas, approach is infeasible

We'll see two more approaches

Today: Tableaux method

Easy to understand, easy to implement

Tableaux

A Tableau is a tree-like structure, where:

- The root is the formula φ we want to check for satisfiability
- Internal nodes contain formulas over propositions of φ
- Leaves are labelled with X or O

The tableau is constructed incrementally

Once construction is complete:

φ is satisfiable iff there exists some leaf labelled with O

equivalently, φ is unsatisfiable iff all leaves are labelled with X

Expansion rules

α -rules (or deterministic rules)

$$\frac{\varphi \wedge \psi}{\varphi}$$

$$\psi$$

$$\frac{\neg(\varphi \vee \psi)}{\neg\varphi}$$

$$\neg\psi$$

$$\frac{\neg(\varphi \supset \psi)}{\varphi}$$

$$\neg\psi$$

$$\frac{\neg\neg\varphi}{\varphi}$$

β -rules (or splitting rules)

$$\frac{\varphi \vee \psi}{\varphi \mid \psi}$$

$$\frac{\neg(\varphi \wedge \psi)}{\neg\varphi \mid \neg\psi}$$

$$\frac{\varphi \supset \psi}{\neg\varphi \mid \psi}$$

$$\frac{\varphi \leftrightarrow \psi}{\varphi \mid \neg\varphi}$$

$$\psi \mid \neg\psi$$

$$\frac{\neg(\varphi \leftrightarrow \psi)}{\neg\varphi \mid \varphi}$$

$$\psi \mid \neg\psi$$

Tableaux Construction Algorithm

Tableaux is constructed proceeding top-down

Start with root labelled with formula to check

At every step one node is expanded (nodes can be expanded only once)

Expansion order is arbitrary (but may affect size of Tableau)

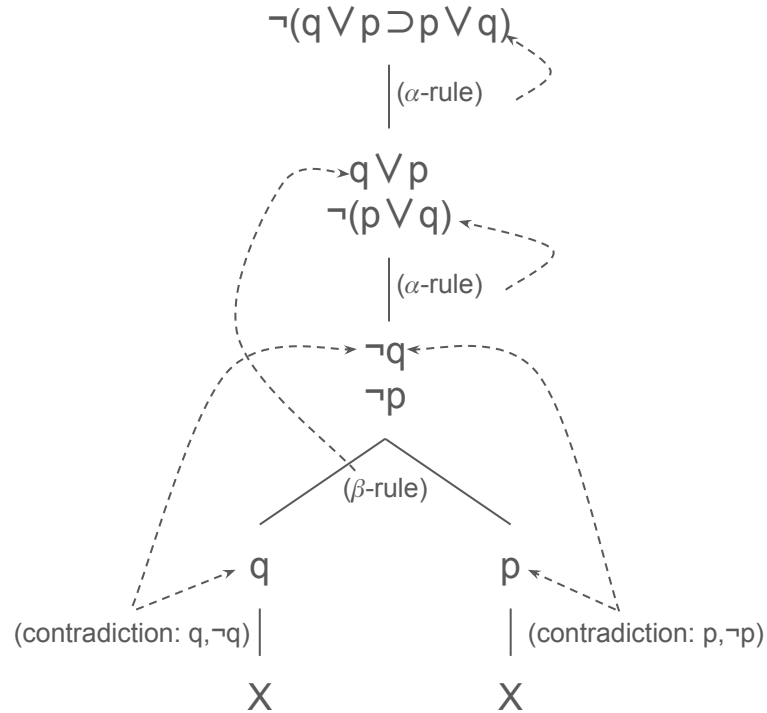
If a path (called *branch*) contains contradictory formulas, e.g., p and $\neg p$, add leaf with X

Once all nodes along a branch have been expanded, if it contains no contradiction then add leaf with O

Construction is complete when every leaf has O or X

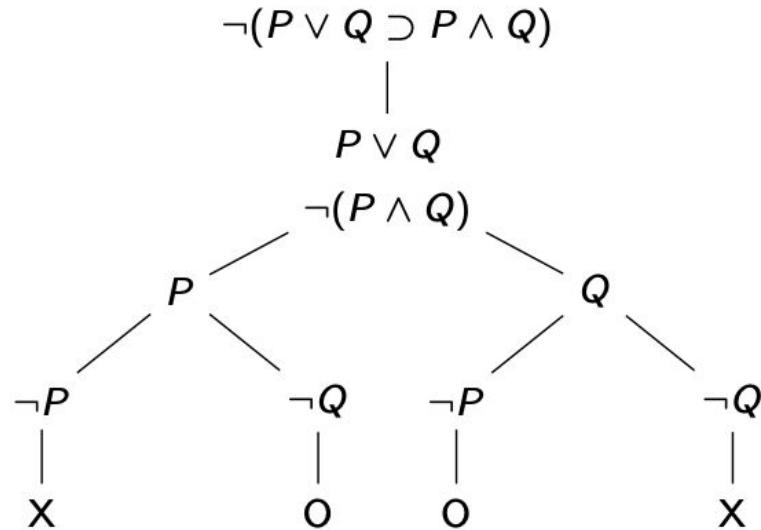
Example

All leaves contain X:
unsatisfiable



Example

Some leaf contains O:
satisfiable



Termination

It is immediate to see that the *tableau construction eventually terminates*:

Applying the expansion rule reduces the size of the formulas involved

Thus, iterative applications of the rules eventually end up in formulas that cannot be expanded anymore

Open and Closed Tableau

A root-leaf path, also called *branch*, is said to be:

- *closed* if it contains a formula and its negation
- *open* otherwise

A tableau is *closed* if all its branches are closed, *open* otherwise

Soundness and Completeness

If a tableau for φ is open then φ is satisfiable (soundness)

If φ is satisfiable, every tableau for φ is open (completeness)

Thus:

φ is satisfiable iff some tableau for φ is open

(Observe: a tableau for φ is open iff all the tableaux for φ are open)

Intuition

Every branch of a tableaux contains, overall, a set of formulas

If an interpretation I satisfies all the formulas in a branch, then $I \models \varphi$

By the construction of the tableau, also the converse holds:

if $I \models \varphi$ then I satisfies all the formulas occurring in some branch of the tableau

Thus,

$I \models \varphi$ iff I satisfies all the formulas in some branch of the tableau

Complexity

Worst-case complexity is $O(2^n)$, for n the size of the formula

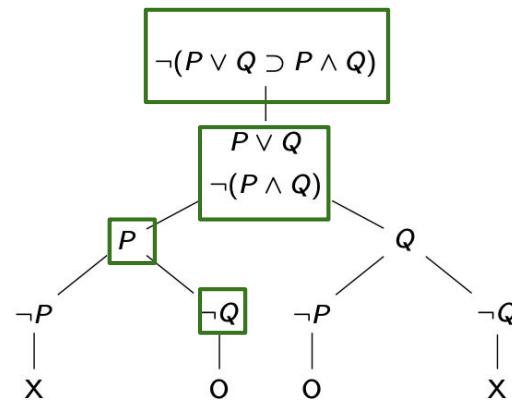
This corresponds to the # of branches produced by the algorithm

(Each branch is of linear size wrt size of the formula)

Models from Tableau

- Select one open branch
- Assign true to every proposition occurring in some *positive literal* (e.g., P) from that branch
- Assign false to every proposition occurring in some *negative literal* (e.g., $\neg Q$) from that branch
- Assign any value to propositions not mentioned in branch
- The resulting set is a model of φ

E.g.: $I=\{P\}$ is a model



Satisfiability, Validity and Logical Implication

Recall the Refutation Principle:

$\Gamma \models \varphi$ iff $\Gamma \cup \{\neg\varphi\}$ is unsatisfiable

If $\Gamma = \{\varphi_1, \dots, \varphi_m\}$ is finite, this is equivalent to:

$\Gamma \models \varphi$ iff $\varphi_1 \wedge \dots \wedge \varphi_m \wedge \neg\varphi$ is unsatisfiable

We can check whether $\Gamma \models \varphi$ by checking whether $\varphi_1 \wedge \dots \wedge \varphi_m \wedge \neg\varphi$ is unsatisfiable

Can use any algorithm for satisfiability, including Tableaux

Special case $\Gamma = \emptyset$: $\models \varphi$ (φ is valid) iff $\neg\varphi$ is unsatisfiable